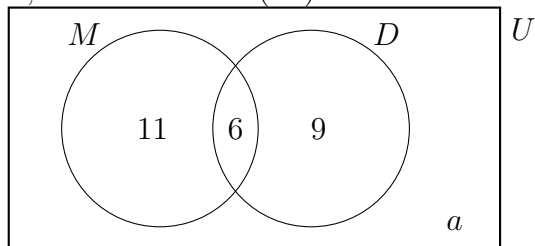


4.19 Quiz: Probability Review

1. In a class of 32 students, some do Music (M) and some do Drama (D).



- (a) Write down the number of students who do Music. [1]
- (b) Find the number of students who do neither Music nor Drama, a . [1]
- (c) One student is chosen at random. Find the probability that
- (i) the student does not do Music; [1]
- (ii) the student does Music or Drama, but not both. [2]
- (d) Given that a student does Drama, find the probability that they also do Music. [2]

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2. The following table shows whether 16 students are in the IB math program and whether they have a TI nSpire calculator.

	nSpire	No nSpire	Total
IB	8	2	10
Non-IB	1	5	6
Total	9	7	16

- (a) Find the probability that a randomly chosen student is in IB and has a nSpire. [1]
- (b) Find $P(\text{nSpire} \mid \text{non-IB})$. [2]
- (c) Determine whether “is in IB” and “has a nSpire” are independent events. Show your working and justify your answer. [3]

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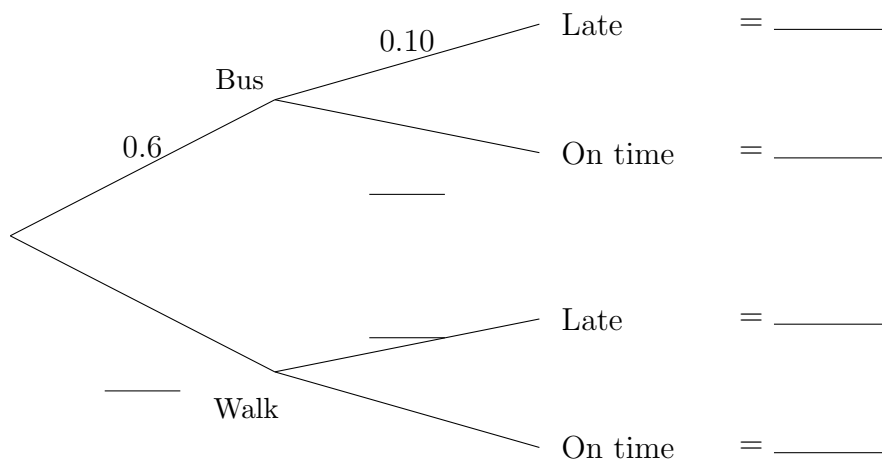
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3. At a school, 60% of students take the bus and 40% walk. Of those who take the bus, 10% arrive late. Of those who walk, 5% arrive late.

(a) Complete the tree diagram below, showing all probabilities. [2]



- (b) Find the probability that a randomly chosen student walks and arrives late. [1]
 (c) Find the probability that a randomly chosen student arrives late. [2]
 (d) Given that a student arrives late, find the probability that they walk. [2]

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4. A bag contains 4 black marbles and 6 white marbles. Two marbles are drawn at random, one after the other, without replacement.
 - (a) Find the probability that both marbles are black. [2]
 - (b) Find the probability that both marbles are the same colour. [3]
 - (c) Find the probability that both marbles are black given that at least one marble is black. [3]